Answer to the Q No-01

Here,

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 3 & 6 & -3 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1 \end{bmatrix}$$

(Answer)

Answer to the Q No-02

Here,

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \begin{bmatrix} R_2' = R_2 - 4R_1 \\ R_3' = R_3 - 7R_1 \end{bmatrix}$$

$$\sim \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} R_3' = R_3 - 2R_2 \end{bmatrix}$$

$$\sim \begin{pmatrix} 3 & 0 & -3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} R_1' = 3R_1 + 2R_2 \end{bmatrix}$$

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$$\sim \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{bmatrix} R_1' = \frac{R_1}{3} \\ R_2' = \frac{R_2}{-3} \end{bmatrix}$$

(Answer)

Answer to the Q No-03

The set of the vectors are,

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ 6 \\ 3 \end{bmatrix}$$

Now,

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$$

$$\Rightarrow \alpha_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha_1 \\ 2\alpha_1 \\ \alpha_1 \end{pmatrix} + \begin{pmatrix} 2\alpha_2 \\ 4\alpha_2 \\ 2\alpha_2 \end{pmatrix} + \begin{pmatrix} 3\alpha_3 \\ 6\alpha_3 \\ 3\alpha_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ 2\alpha_1 + 4\alpha_2 + 6\alpha_3 \\ \alpha_1 + 2\alpha_2 + 3\alpha_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Thus,

$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = 0$$
$$2\alpha_1 + 4\alpha_2 + 6\alpha_3 = 0$$
$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = 0$$

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$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 4 & 6 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - R_1 \end{bmatrix}$$

Here, we can see that the vectors are linearly dependent.

From the first equation we get,

$$\alpha_1 = -2\alpha_2 - 3\alpha_3$$

Let, $\alpha_2 = 1$ and $\alpha_3 = 0$ Then,

$$\alpha_1 = -2(1) - 3(0)$$
$$= -2$$

$\therefore -2v_1 + v_2 = 0$

Let, $\alpha_2 = 0$ and $\alpha_3 = 1$ Then,

$$\alpha_1 = -2(0) - 3(1)$$
$$= -3$$

$\therefore -3v_1 + v_3 = 0$

(Answer)

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Answer to the Q No-04

Here,

$$b=\begin{bmatrix}1\\2\\3\end{bmatrix},\quad u=\begin{bmatrix}1\\0\\1\end{bmatrix},\quad v=\begin{bmatrix}0\\1\\1\end{bmatrix}$$

Now,

$$\alpha_1 u+\alpha_2 v=b$$

$$\Rightarrow \alpha_1\begin{pmatrix}1\\0\\1\end{pmatrix}+\alpha_2\begin{pmatrix}0\\1\\1\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\alpha_1\\0\\\alpha_1\end{pmatrix}+\begin{pmatrix}0\\\alpha_2\\\alpha_2\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_1+\alpha_2\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}$$

So,

$$\alpha_1=1$$

$$\alpha_2=2$$

$$\alpha_1+\alpha_2=3$$

Thus,

$$b=u+2v$$

(Answer)

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Answer to the Q No-05

Here,

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Now,

$$\alpha_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha_1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \alpha_2 \\ 0 \end{pmatrix} + \begin{pmatrix} \alpha_3 \\ \alpha_3 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha_1 + \alpha_3 \\ \alpha_2 + \alpha_3 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

So,

$$\alpha_1 + \alpha_3 = a$$
$$\alpha_2 + \alpha_3 = b$$
$$0 = c$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 1 & a \\ 0 & 1 & 1 & b \\ 0 & 0 & 0 & c \end{array} \right)$$

Since the z component does not have a pivot, the set S does not span $\mathbb{R}^3$.

(Answer)

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Answer to the Q No-06

Here,

$$T = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \mid x+2y-z=0 \right\}$$

Now,

A4: There exists a $\vec{0} \in V$ such that $u \oplus \vec{0} = u$ for all $u \in V$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Here,

$$0+2.0-0=0$$

Thus, A4 holds.

A1: $u \oplus v \in V$ for all $u, v \in V$

Let,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} ; x+2y-z=0$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} ; a+2b-c=0$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a+x \\ b+y \\ c+z \end{pmatrix}$$

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Here,

$$(a+x)+2(b+y)-(c+z)$$

$$= a+x+2b+2y-c-z$$

$$=(a+2b-c)+(x+2y-z)$$

$$=0+0$$

$$=0$$

Thus, A_1 holds.

<u>M_1:</u> $k.u \in V$ for all $u \in V$ and $k \in F$

Now,

$$k\begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$=\begin{pmatrix} kx \\ ky \\ kz \end{pmatrix}$$

Here,

$$kx+2ky-kz$$

$$=k(x+2y-z)$$

$$=k.0$$

$$=0$$

Thus, M_1 also holds, proving that the given set T is a subspace of $\mathbb{R}^3$.

(Answer)

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Answer to the Q No-07

Here,

$$W = \{a+bx \mid a, b \in \mathbb{R}\}$$

Now,

A4: There exists a $\vec{0} \in V$ such that

$$u \oplus \vec{0} = u \text{ for all } u \in V$$

$$(a+bx) + (0) = (a+bx)$$

Here,

$$0 + 0.x = 0 \quad [a=0, b=0]$$

Thus, A4 holds.

A_1: $u \oplus v \in V$ for all $u, v \in V$

Let,

$$a_1 + b_1 x$$

$$a_2 + b_2 x$$

$$(a_1 + b_1 x) + (a_2 + b_2 x)$$

$$= (a_1 + a_2) + (b_1 + b_2) x$$

Here, $(a_1 + a_2), (b_1 + b_2) \in \mathbb{R}$:

Thus, A_1 holds.

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M_1 : $k \cdot u \in V$ for all $u \in V$ and $k \in F$

$$k(a+bx)$$

$$= ka+kbx$$

Here, $ka, kb \in \mathbb{R}$

Thus, M_1 also holds, proving that W is a subspace of V.

(Answer)

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Answer to the Q No-08

Here,

$$C=\begin{pmatrix}1 & -1 & 2\\ 2 & -2 & 4\\ -1 & 1 & -2\end{pmatrix}$$

Now,

$$\begin{pmatrix}1 & -1 & 2\\ 2 & -2 & 4\\ -1 & 1 & -2\end{pmatrix}\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\left(\begin{array}{ccc|c}1 & -1 & 2 & 0\\ 2 & -2 & 4 & 0\\ -1 & 1 & -2 & 0\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}1 & -1 & 2 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{array}\right)\begin{bmatrix}R_2'=R_2-2R_1\\ R_3'=R_3+R_1\end{bmatrix}$$

Here, y and z are free variables.

Thus, $y=t_1$

$z=t_2$

$x-y+2z=0$

$\Rightarrow x-t_1+2t_2=0$

$\Rightarrow x=t_1-2t_2$

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Now,

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} t_1 - 2t_2 \\ t_1 \\ t_2 \end{pmatrix}$$

$$= t_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$$

$\therefore$ Null space = span $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\}$

$\therefore$ Basis = $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\}$

Nullity = 2

Dimension = 2

(Answer)

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Answer to the Q No-09

Here,

$$V_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, V_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 8 \end{bmatrix}, V_3 = \begin{bmatrix} 3 \\ 6 \\ 9 \\ 12 \end{bmatrix}$$

Now,

$$\alpha_1 \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \alpha_2 \begin{pmatrix} 2 \\ 4 \\ 6 \\ 8 \end{pmatrix} + \alpha_3 \begin{pmatrix} 3 \\ 6 \\ 9 \\ 12 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha_1 + 2\alpha_2 + 3\alpha_3 \\ 2\alpha_1 + 4\alpha_2 + 6\alpha_3 \\ 3\alpha_1 + 6\alpha_2 + 9\alpha_3 \\ 4\alpha_1 + 8\alpha_2 + 12\alpha_3 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

So, $\alpha_1 + 2\alpha_2 + 3\alpha_3 = a$

$2\alpha_1 + 4\alpha_2 + 6\alpha_3 = b$

$3\alpha_1 + 6\alpha_2 + 9\alpha_3 = c$

$4\alpha_1 + 8\alpha_2 + 12\alpha_3 = d$

$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 2 & 4 & 6 & b \\ 3 & 6 & 9 & c \\ 4 & 8 & 12 & d \end{array}\right)$$

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$$\sim \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 0 & 0 & b-2a \\ 0 & 0 & 0 & c-3a \\ 0 & 0 & 0 & d-4a \end{array}\right) \begin{bmatrix} R_2' = R_2 - 2R_1 \\ R_3' = R_3 - 3R_1 \\ R_4' = R_4 - 4R_1 \end{bmatrix}$$

Since, the y and z components do not contain any pivots. and if $b = 2a$, $c = 3a$ and $d = 4a$ then the basis will be,

$$\text{Basis} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \right\}$$

(Answer)

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Answer to the Q No-10

Here,

$$U = \left\{ \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} \in \mathbb{R}^4 \mid x-y+z=0 \text{ and } y+w=0 \right\}$$

Now,

A4: There exists a $\vec{0} \in V$ such that $u \oplus \vec{0} = u$ for all $u \in V$

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$$

Here,

$$0-0+0=0$$
$$0+0=0$$

Thus, A4 holds.

A1: $u \oplus v \in V$ for all $u, v \in V$

Let,

$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} ; \quad \begin{matrix} x-y+z=0 \\ y+w=0 \end{matrix}$$

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} ; \quad \begin{matrix} a-b+c=0 \\ b+d=0 \end{matrix}$$

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$$\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} + \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} a+x \\ b+y \\ c+z \\ d+w \end{pmatrix}$$

Here,

$(a+x) - (b+y) + (c+z)$

$= a+x-b-y+c+z$

$= (x-y+z) + (a-b+c)$

$= 0+0$

$= 0$

And,

$(b+y) + (d+w)$

$= (y+w) + (b+d)$

$= 0+0$

$= 0$

Thus, A_1 holds.

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M_1: $k.u \in V$ for all $u \in V$ and $k \in F$

Now,

$$k\begin{pmatrix} x \\ y \\ z \\ \omega \end{pmatrix}$$

$$= \begin{pmatrix} kx \\ ky \\ kz \\ k\omega \end{pmatrix}$$

Here,

$$kx - ky + kz$$

$$= k(x - y + z)$$

$$= k.0$$

$$= 0$$

And,

$$ky + k\omega$$

$$= k(y + \omega)$$

$$= k.0$$

$$= 0$$

Thus, M_1 also holds, proving that U is a subspace of $\mathbb{R}^4$.

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Now,

$$x - y + z = 0$$

$$y + w = 0$$

$$\begin{pmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \quad [R_1' = R_1 + R_2]$$

Here, z and w are free variables. Thus,

$$z = t_1$$

$$w = t_2$$

$$x + t_1 + t_2 = 0$$

$$\Rightarrow x = -t_1 - t_2$$

$$y + t_2 = 0$$

$$\Rightarrow y = -t_2$$

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So,

$$\begin{pmatrix} x \\ y \\ z \\ \omega \end{pmatrix} = \begin{pmatrix} -t_1 - t_2 \\ -t_2 \\ t_1 \\ t_2 \end{pmatrix}$$

$$= t_1 \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, t_2 \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\therefore \text{Basis} = \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

(Answer)

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